Spiny dogfish (*Squalus acanthias*)

Summary Scores

- **Sensitivity:** Low
- **Exposure:** Moderate
- **Vulnerability:** Low

Sensitivity

Sensitivity Attributes	Score	LH DQ	Tally
Adult Mobility	1.16	2.6	
Complexity in Reproductive Strategy	1.48	2.6	
Dispersal of Early Life Stages	1.24	3	
Egg and Larval Survival and Settlement Requirements	1	3	
Historical Variability in Temperature	1.8	2.8	
Juvenile Habitat Requirements	1.6	2.5	
Other Stressors	1.8	2	
Population Growth Rate	3.56	2.75	
Prey Specificity	1.16	3	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	1.32	2.2	
Spawning Cycle	1.56	2.6	
Stock Size/Status	2	3	
Total Sensitivity | Certainty	**Low (1) | 1**		

Exposure

Exposure Factors	Mean	SD	Tally
Bottom Temperature	2.36	1.10	
Bottom Oxygen	2.67	1.00	
Bottom Salinity	2.75	0.99	
Bottom Aragonite Solubility	1.31	0.77	
Small zooplankton integrated biomass	1.31	0.35	
Medium zooplankton integrated biomass	1.36	0.32	
Large zooplankton integrated biomass	1.24	0.33	
Total Exposure	**Moderate (2)**	**0.94**	

Vulnerability

Vulnerability	Mean	SD	Tally
Total Vulnerability	**Low (2)**	**0.94**	

Low Moderate High Very High

Figure 1: Summary Sensitivity, Exposure, and Vulnerability Scores and Score Distributions for Spiny dogfish. Exposure and Vulnerability scores are averaged spatially across the species distribution and within stocks if applicable (see Figure 5 and Figure 4). Life History Data Quality (LH DQ) and Standard Deviations (SD) are reported for senstivity and exposure/vulnerability scores, respectively.

Changes from Previous Assessment(s)

Table 1: Changes to Vulnerability, Exposure, and Sensitivity scores relative to Previous Assessments. If a citation for the previous assessment is not provided, it comes from Hare et al. (2016).

Metric	Previous Assessment(s)	Current Assessment	Change
Sensitivity	Low	Low	None
Exposure	High	Moderate	Decrease
Vulnerability	Low	Low	None
Directional Effect	Neutral	Neutral	None

Stocks

Spiny dogfish is managed as a single stock or stock information was not considered for this iteration of the CVA.

Score Descriptions:

Sensitivity:

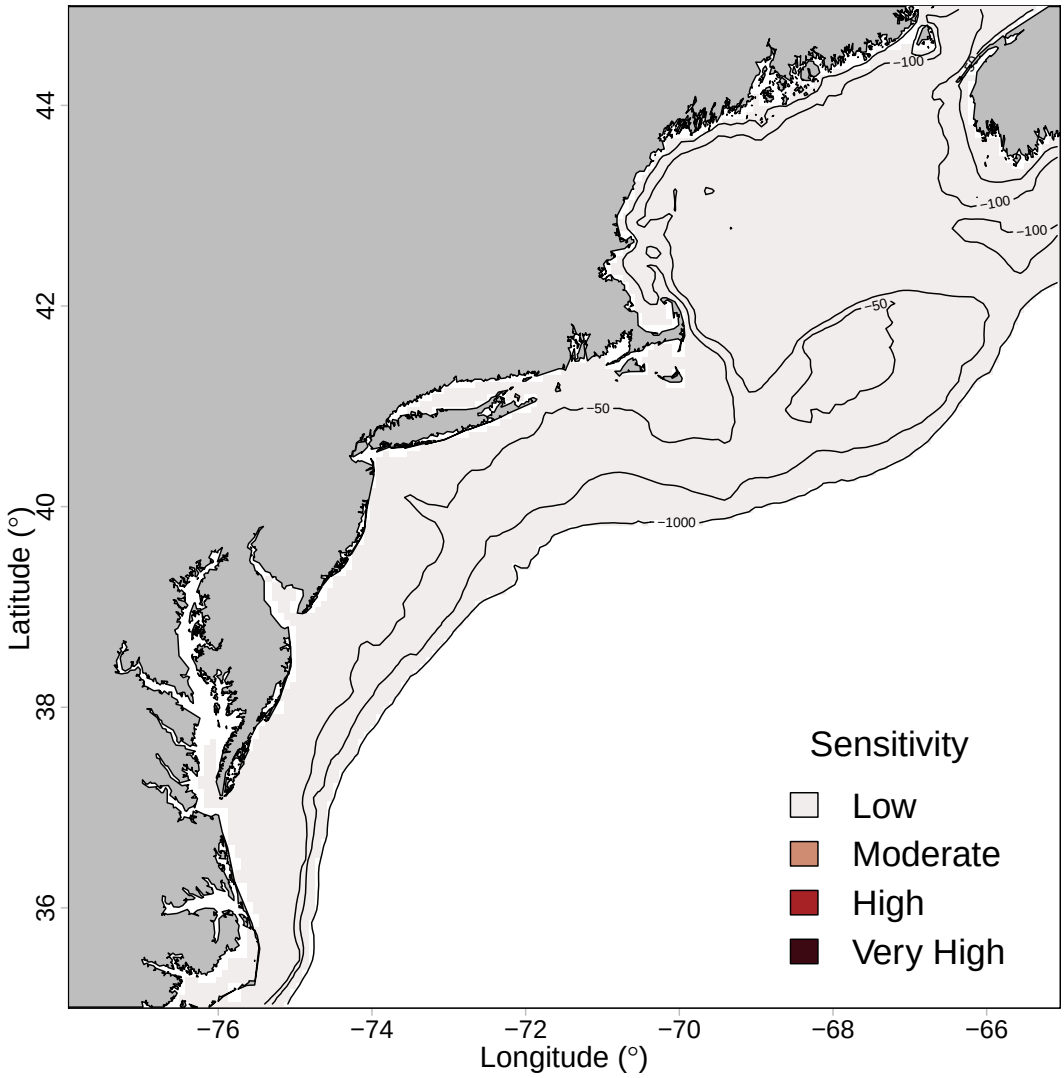


Figure 2: Spiny dogfish Sensitivity on the Northeast Shelf

Spiny dogfish received a **Low** sensitivity score, with a certainty of 100%. The only attribute with a score greater than 2.5 was Population Growth Rate (3.6) (Figure 1). They are a long-lived, slowly maturing species. Age at maturity ranges from 6 years for males and 12 years for females, with maximum lifespans of 35-40 years (Stehlik 2007). Gestation periods last from 18-24 months (Stehlik 2007).

Spiny dogfish also received a Low sensitivity score in the previous Climate Vulnerability Assessment. This score was also driven by Population Growth Rates (Hare et al. 2016).

Exposure:

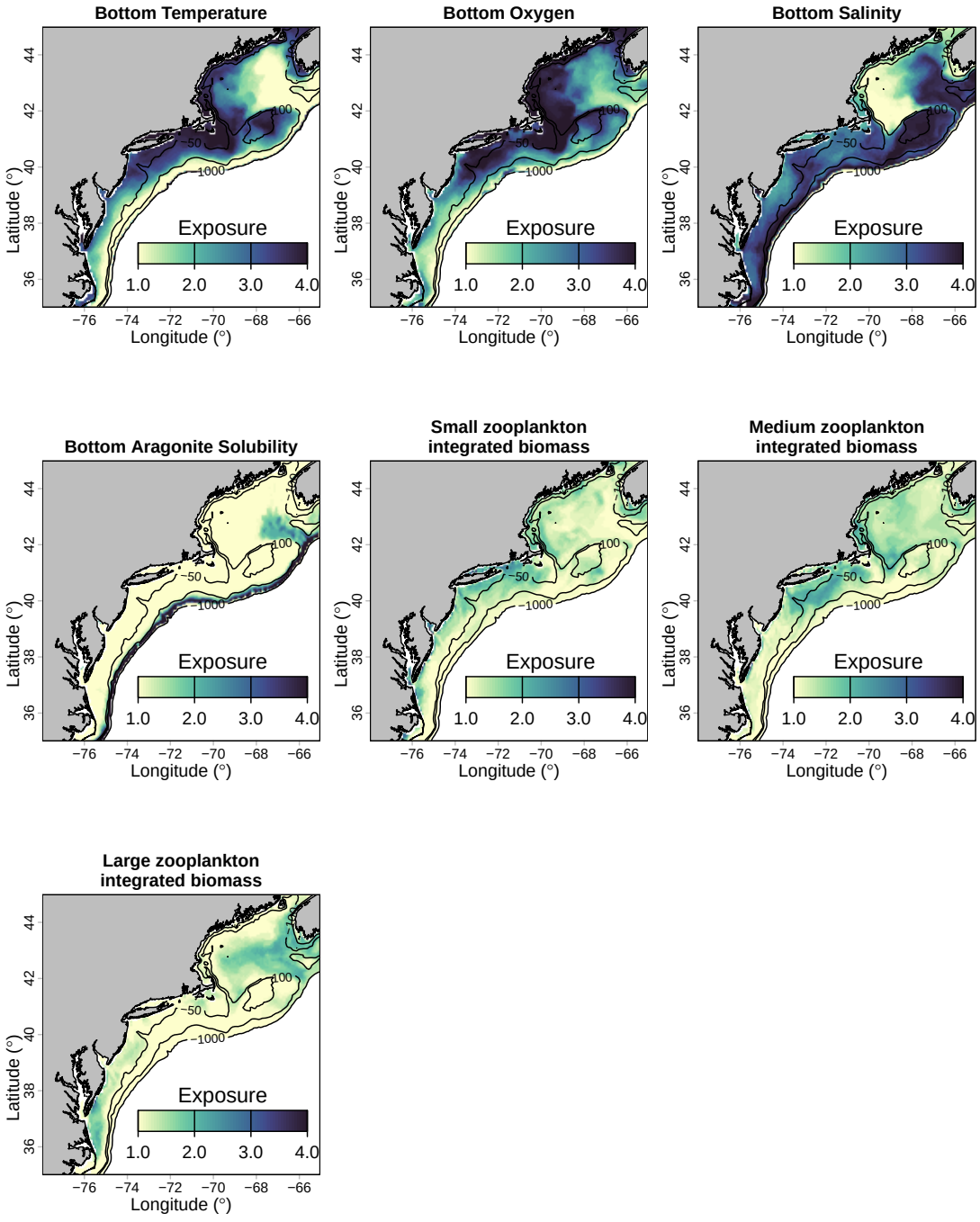


Figure 3: Maps of each Exposure Factor for Spiny dogfish. Exposure was calculated between 2014-2023 and 2025-2035

Mean Exposure scores for Spiny dogfish were **Moderate** (2 +/- 0.94) (Figure 1). Exposure was Very High over Georges Bank and High along the coast of the Gulf of Maine and in Southern New England.

Scores were also High on the continental shelf break (Figure 4). These patterns were driven by bottom temperature, oxygen, salinity, and aragonite saturation scores (Figure 3).

In the previous Assessment, spiny dogfish received a High Exposure score, driven by high sea surface temperature and ocean acidification exposure scores (Hare et al. 2016).

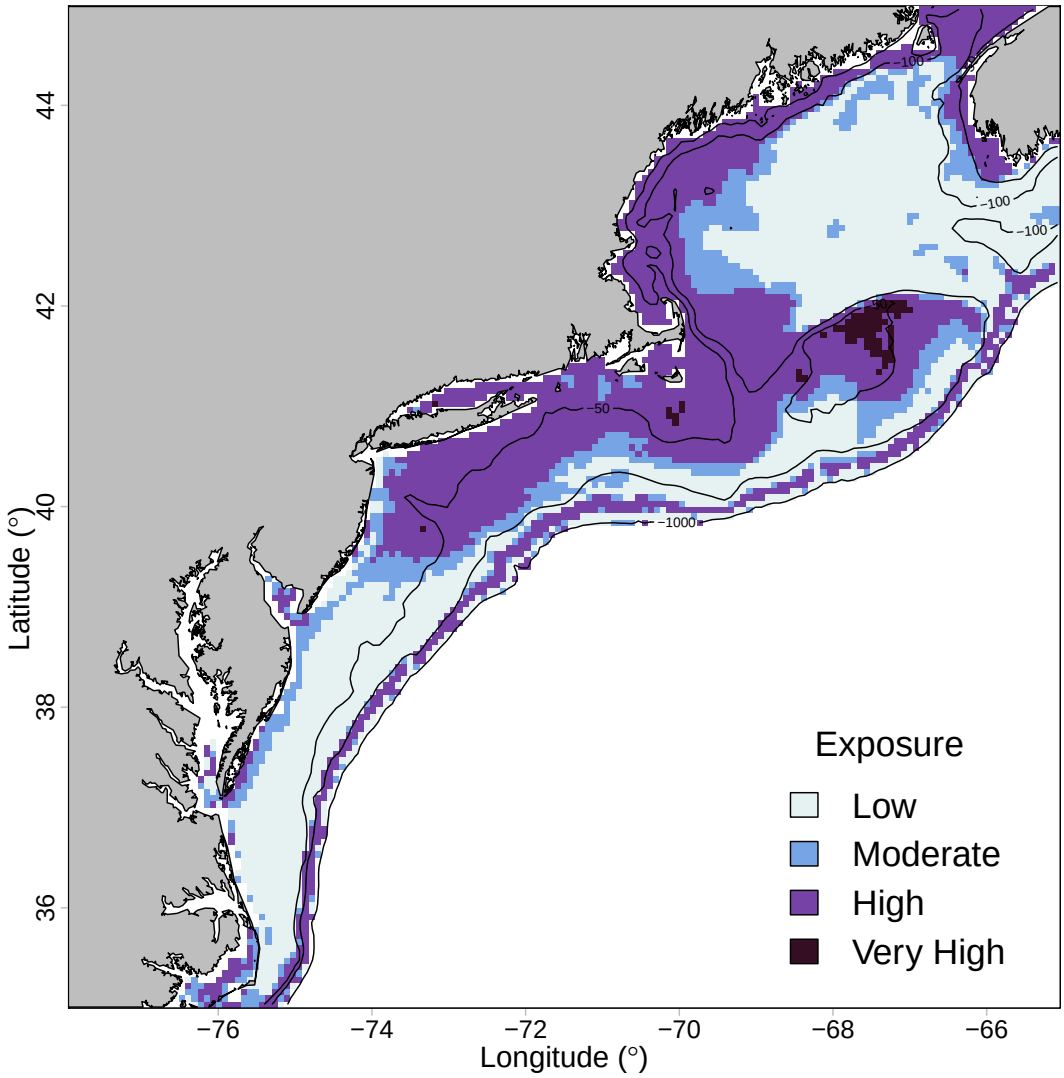


Figure 4: Total Exposure on the Northeast Shelf for Spiny dogfish. Total Exposure is calculated by applying the logic rule from Hare et al. (2016) within grid cells across exposure factors (see Figure 3)

Vulnerability:

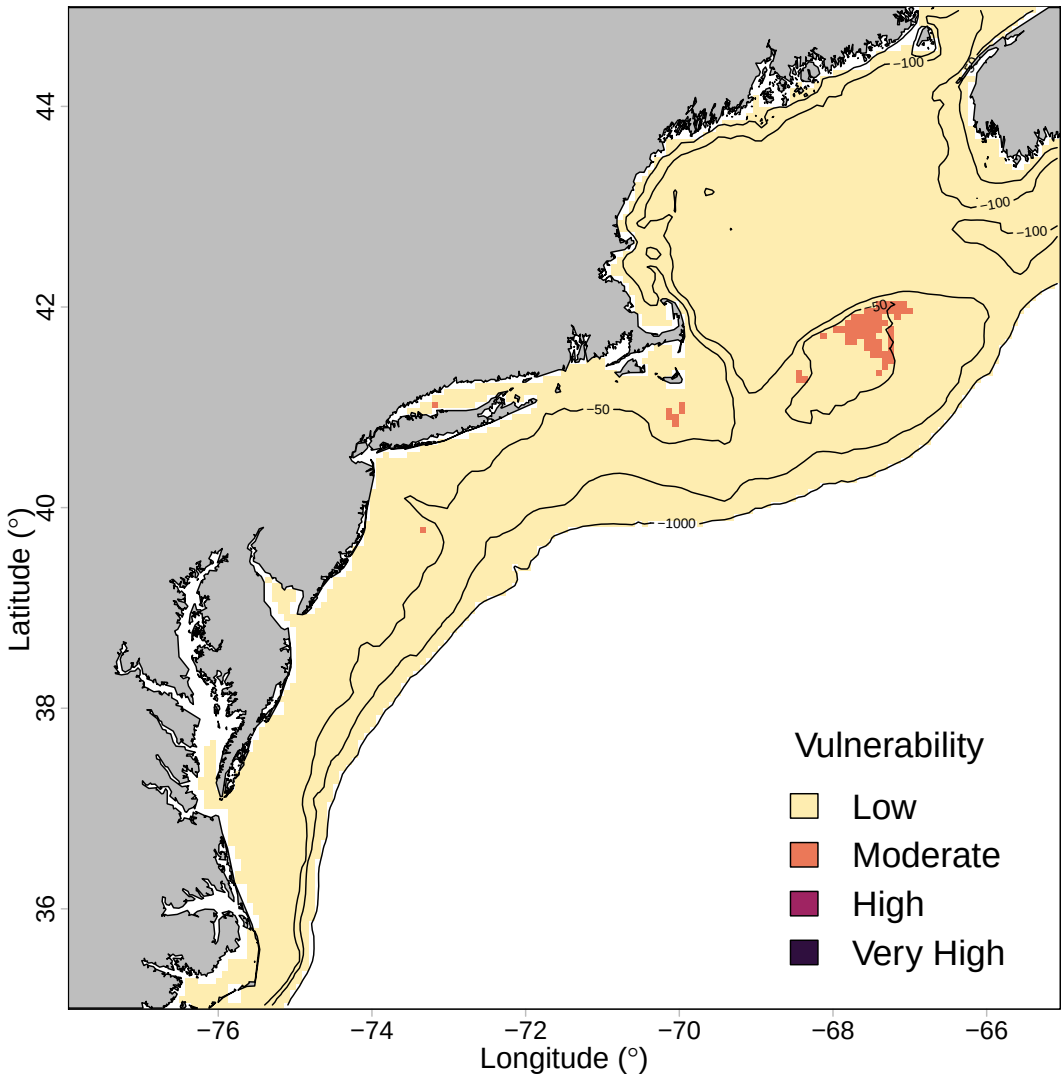


Figure 5: Vulnerability on the Northeast Shelf for Spiny dogfish. Vulnerability is the product of sensitivity and exposure (see Figure 4 and Figure 2)

Mean Vulnerability scores for Spiny dogfish were Low (2 +/- 0.94) (Figure 1). Exposure was mostly uniform across the model species range. There were some Moderate scores over Georges Bank (Figure 5).

In the previous Assessment, spiny dogfish also received a Low Vulnerability score (Hare et al. 2016).

Data Quality Metrics

Life History Data Quality:

90% of data quality scores were greater than or equal to 2, indicating excellent data quality. No data gaps (mean attribute data quality score < 2) were identified (Figure 1).

Model Confidence Score:

The ensemble spiny dogfish model received a mean model confidence score of 0.9 (+/- 0.3), indicating no confidence in the model. The predicted distributions did not meet experts' expectations. Experts suggested that the model may be overfit. The ensemble model AUC score was 0.77, indicating fair model performance.

Additional Metrics

Distribution Shifts:

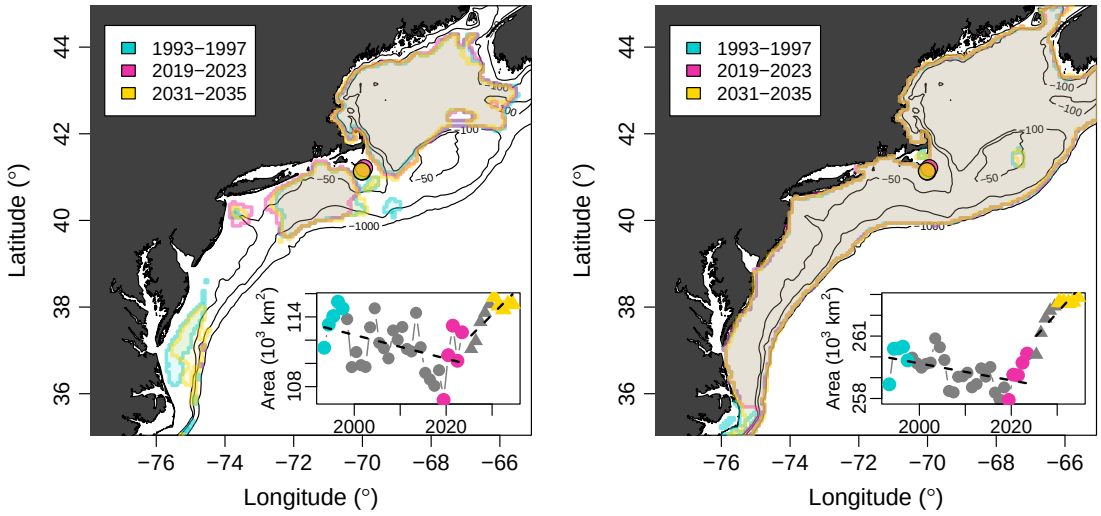


Figure 6: Distribution shifts on the Northeast Shelf for Spiny dogfish. Polygons illustrate core habitat (left) and species range (right) for five-year average periods. Points illustrate weighted centroid, calculated as the average position with species distribution models as weights. Inset timeseries illustrate the shift in core and habitat range area over time. Circles represent points from the MOM6 hindcast; triangles represent points from the MOM6 forecast. Dashed lines represent significant linear regressions

Note: For an animated version of this plot, please see the online version of this document.

Spiny dogfish core habitat and range shrunk significantly between 1993 and 2023. Spiny dogfish lost core habitat off the coast of Virginia and the southern edge of their range moved northward during this time period (Figure 6). The weighted centroid shifted 10 km north-northeast.

Under near future ocean conditions, spiny dogfish core habitat and range are predicted to expand to near 1993 distributions, including reestablishing core habitat off the coast of Virginia (Figure 6). The weighted centroid is predicted to shift 10 km back southwest.

In the previous assessment, spiny dogfish received a Very High Distributional Vulnerability Rank due to their habitat preferences and high mobility (Hare et al. 2016).

Table 2: Distribution shifts based on five-year average distributions. Changes are calculated relative to the previous time period.

Time Period	Weighted Centroid Shift (km & bearing)	Core Habitat Area (change) (km^2)	Habitat Total Area (change) (km^2)	Latitude Range (change) (deg)	Change in Longitude Range (deg)
1993-1997	NA	114,805	260,321	7.05	9.14
2019-2023	10.2 km NNE	110,812 (-3,993)	259,274 (-1,047)	6.86 (-0.19)	9.06 (-0.08)
2031-2035	9.7 km SW	115,098 (+4,287)	262,657 (+3,383)	7.05 (+0.19)	9.14 (+0.08)

Directional Effect:

Experts predict that climate change will have a **neutral** effect on spiny dogfish. Certainty was 91%. Occurrence of spiny dogfish is correlated with warmer temperatures (Sagarese, Frisk, Cerrato, et al. 2014). They are highly migratory and will move north-south to follow waters within their preferred temperature range (6-13C) (Stehlik 2007) (Carlson et al. 2014). They may benefit from the loss of cod as they occupy similar feeding niches (Neuenhoff et al. 2019) (Ono et al. 2019). The previous Assessment also predicted that climate change would have a neutral impact on spiny dogfish (Hare et al. 2016).

Climate Effects on Abundance and Distribution

Nye et al. (2009) observed a range expansion and southward shift in center of biomass in spiny dogfish, although changes in biomass over the study period may have influenced these shifts. Seasonal occurrence of spiny dogfish is highly correlated with depth and bottom temperature, with a higher probability of occurrence in warmer years versus colder years (Sagarese, Frisk, Cerrato, et al. 2014). Spiny dogfish shift their distribution seasonally; variation in environmental drivers may impact population and fishery dynamics as temperatures continue to increase. Climate-inducing seasonal distribution shifts influencing habitat availability for spiny dogfish may also lead to vulnerability of the inshore fishery in the fall (Sagarese, Frisk, Miller, et al. 2014).

Life History Synopsis

Atlantic spiny dogfish are a schooling, highly migratory shark species (HMS) inhabiting inshore and offshore waters from Labrador to Florida, although they are most abundant from Nova Scotia and Cape Hatteras. Spiny dogfish perform diel vertical migrations; most schools are found around 40-50 m depth, but can descend upwards of 500 m (Stehlik 2007) (Carlson et al. 2014). There are populations that occupy the continental shelves of Europe, Asia, and the northeast Pacific, but these populations do not mix (Stehlik 2007). Spiny dogfish perform north-south migrations and move between inshore-offshore waters in response to water temperature, migrating north in the spring and summer and south in the fall and winter to remain in waters between 6-13°C (Stehlik 2007) (Carlson et al. 2014). Age at maturity is 6 years for males and 12 years for females, with a lifespan of 35 years for males and 40 years for females. Spawning occurs offshore in the winter, with a gestation period of 18-24 months (Stehlik 2007). Proportions of mature females in catch are correlated with bottom temperatures, with more mature females caught at relatively higher bottom temperatures. However, how temperature impacts reproduction is still unclear (Sagarese, Frisk, Miller, et al. 2014). Spiny dogfish are opportunistic feeders, preying upon a variety

of fishes (herring, mackerel, hakes, sand lance, menhaden), squid, ctenophores, polychaetes, crustaceans, bivalves. Diets can shift based on location and availability (Stehlik 2007) (DeVries, Gartland, and Latour 2025). Predators of spiny dogfish include sharks, seals, cod, goosefish, dolphins, and larger fish (Stehlik 2007). Spiny dogfish occupy a similar niche to that of Atlantic cod, whose populations have been impacted by increasing seal populations in the Northwest Atlantic and Canada. It has been hypothesized that the loss of cod could be a benefit to spiny dogfish, since they occupy similar feeding niches (Neuenhoff et al. 2019) (Ono et al. 2019). The Mid-Atlantic Fishery Management Council (MAFMC) leads the joint management of the Spiny Dogfish Fishery Management Plan, with support from the New England Fishery Management Council (NEFMC) and Atlantic States Marine Fisheries Commission (ASMFC) (Mid-Atlantic Fishery Management Council (MAFMC) 2025). The ASMFC manages spiny dogfish in state waters via the Interstate Fishery Management Plan for Spiny Dogfish (Atlantic States Marine Fisheries Commission (ASMFC) 2002). As of 2023, spiny dogfish in the Northwest Atlantic are not overfished and overfishing is not occurring (Northeast Fisheries Science Center (NEFSC) 2023).

Supplemental Scores

Exposure Timeseries

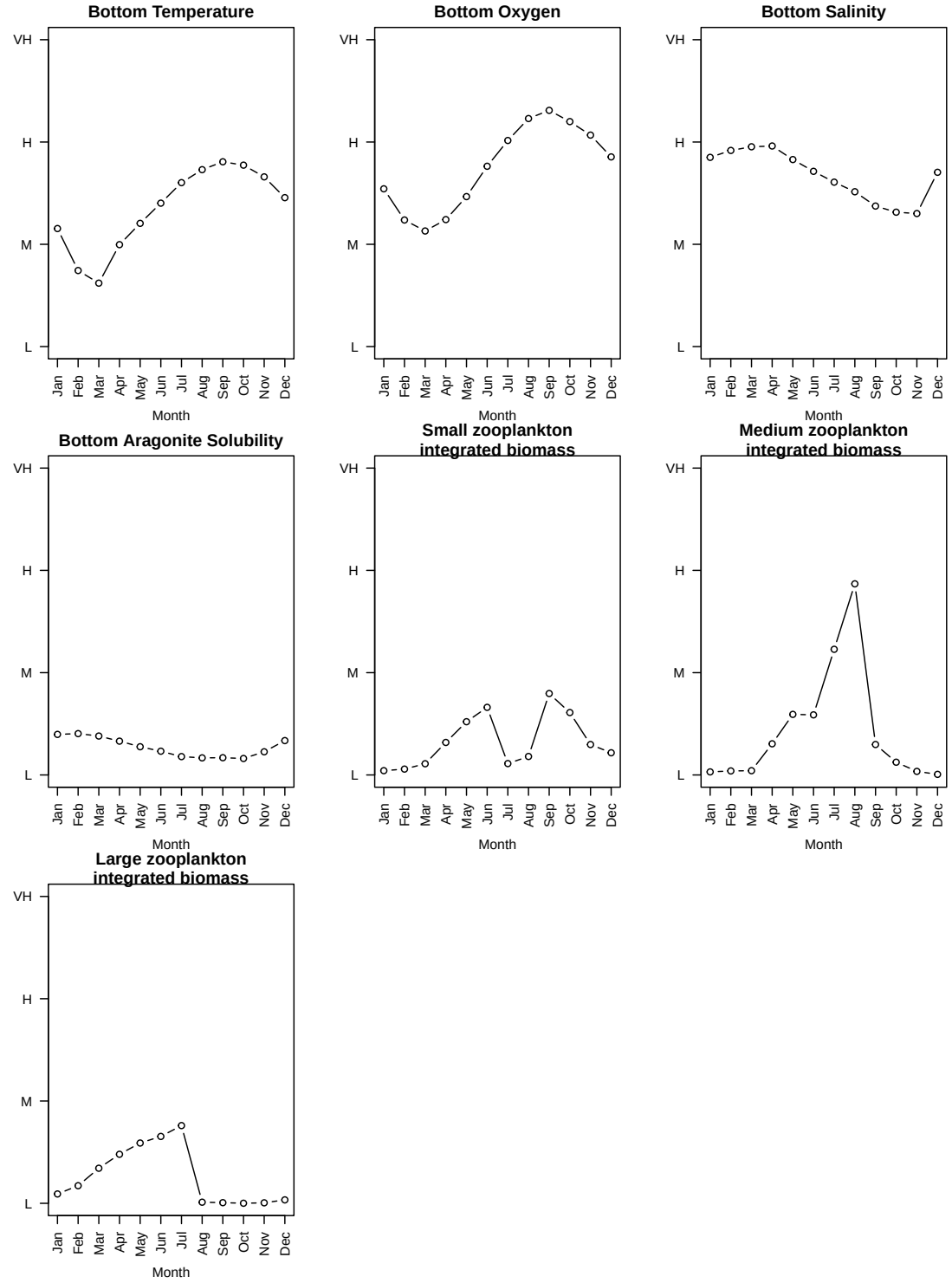


Figure 7: Monthly Timeseries of each Exposure Factor for Spiny dogfish. Exposure was calculated between 2014-2023 and 2025-2035

There was little monthly variation in Exposure scores for spiny dogfish (Figure 8). Scores ranged from Low to Moderate and were lowest from February through May due to low bottom temperature and oxygen exposure scores (Figure 7).

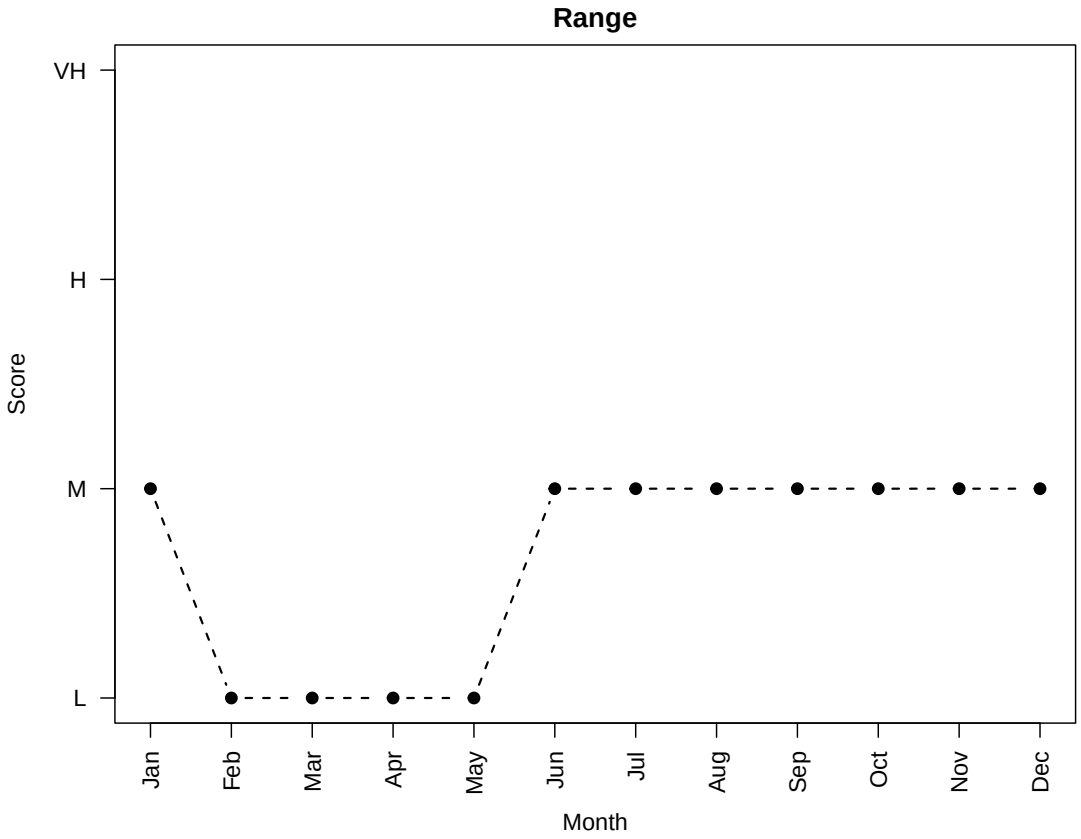


Figure 8: Timeseries of Total Exposure for Spiny dogfish

Exposure due to Significant Factors

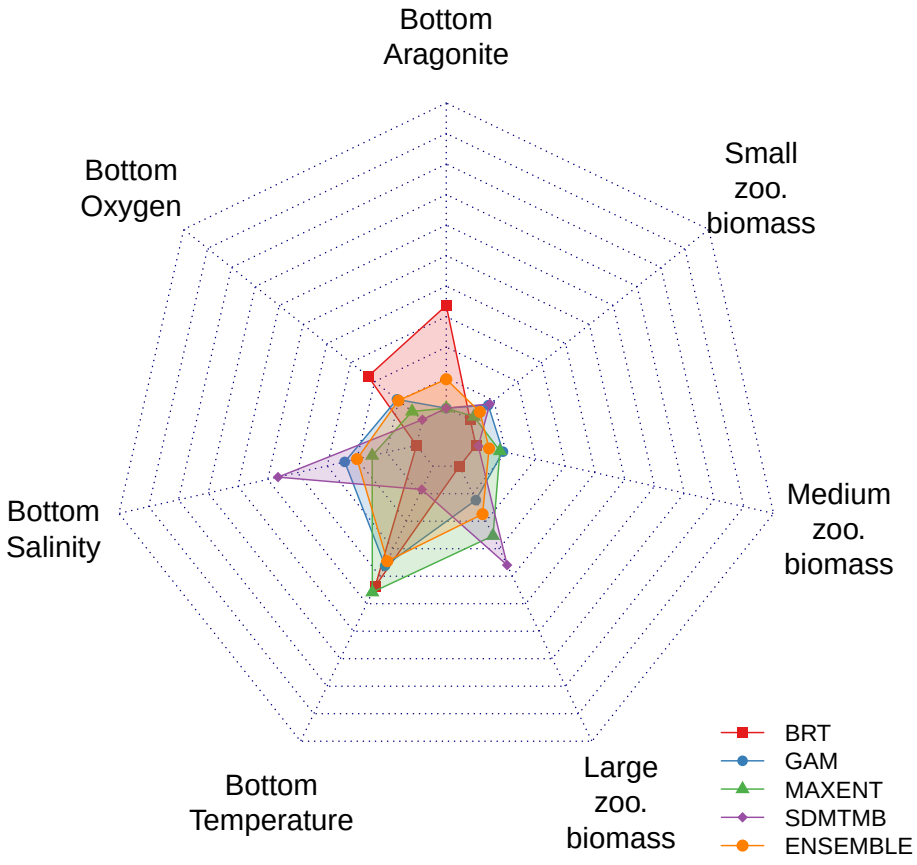


Figure 9: Dynamic Factors driving Spiny dogfish distributions. The closer the polygon is to the edge of the figure, the more important the variable is. Each ring represents 10% variance explained. Factors must explain at least 10% of variance in the ensemble model to be used to calculate total exposure.

Bottom temperature, oxygen, and salinity, as well as large zooplankton biomass were important dynamic drivers of spiny dogfish distributions (Figure 9). Temperatures drive migration patterns (Stehlik 2007) (Carlson et al. 2014). Spiny dogfish are also generalist piscivores (Stehlik 2007) (DeVries, Gartland, and Latour 2025). When exposure was calculated using only these important factors, scores remained Moderate and distributions of scores remained relatively similar (Figure 11; Figure 10).

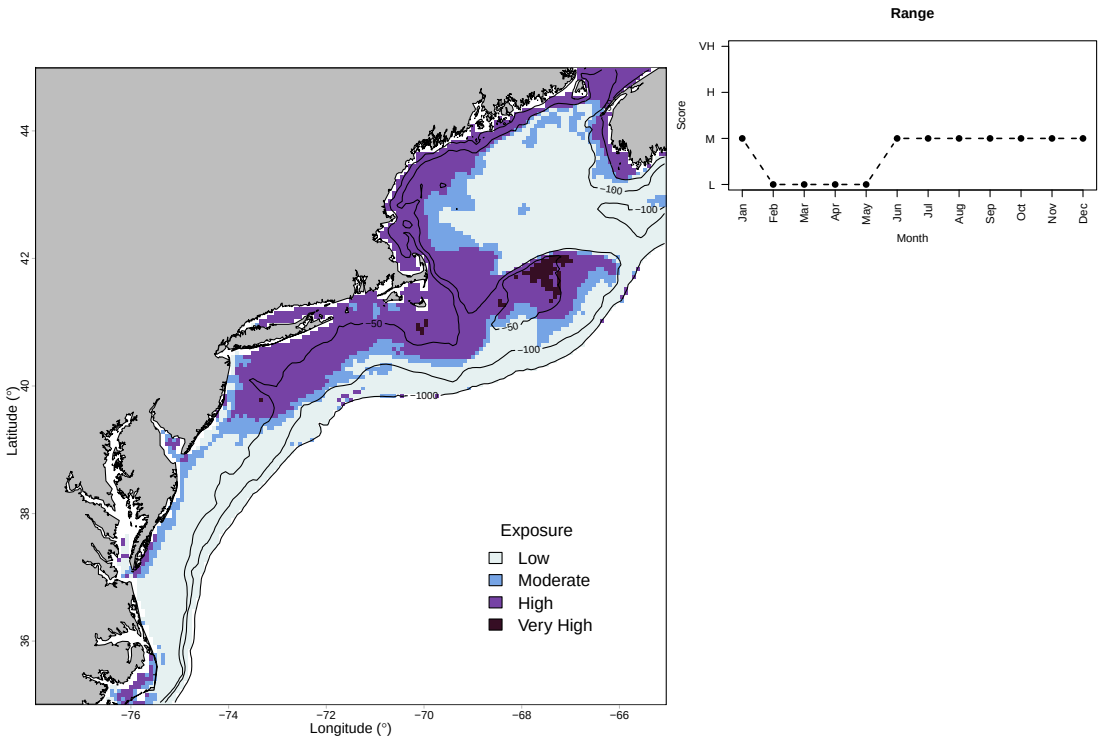
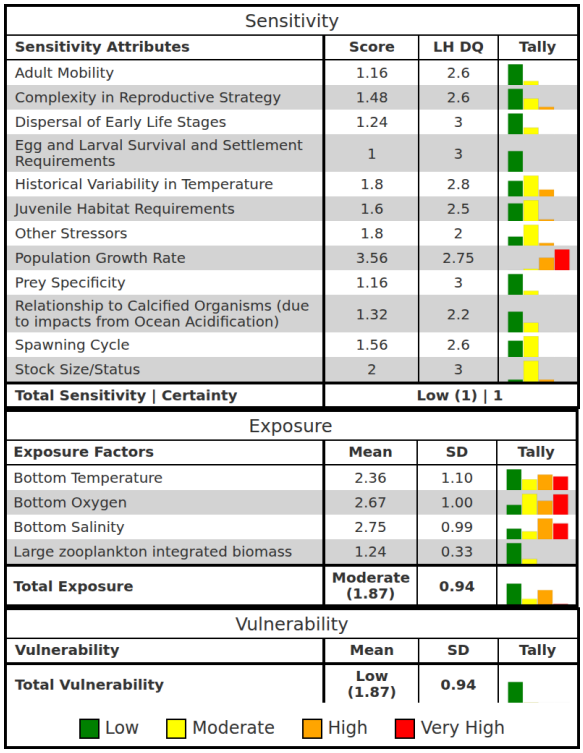


Figure 10: Total Exposure Map and Timeseries when Exposure was calculated with only Exposure Factors critical to Spiny dogfish distributions



Sensitivity

Sensitivity Attributes	Score	LH DQ	Tally
Adult Mobility	1.16	2.6	
Complexity in Reproductive Strategy	1.48	2.6	
Dispersal of Early Life Stages	1.24	3	
Egg and Larval Survival and Settlement Requirements	1	3	
Historical Variability in Temperature	1.8	2.8	
Juvenile Habitat Requirements	1.6	2.5	
Other Stressors	1.8	2	
Population Growth Rate	3.56	2.75	
Prey Specificity	1.16	3	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	1.32	2.2	
Spawning Cycle	1.56	2.6	
Stock Size/Status	2	3	
Total Sensitivity | Certainty	**Low (1) | 1**		

Exposure

Exposure Factors	Mean	SD	Tally
Bottom Temperature	2.36	1.10	
Bottom Oxygen	2.67	1.00	
Bottom Salinity	2.75	0.99	
Large zooplankton integrated biomass	1.24	0.33	
Total Exposure	**Moderate (1.87)**	**0.94**	

Vulnerability

Vulnerability	Mean	SD	Tally
Total Vulnerability	**Low (1.87)**	**0.94**	

Figure 11: As in Figure 1 for Spiny dogfish, but with only significant Exposure factors and the resulting Vulnerability Scores.

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